

7.4.3 Magnetic Flux (ψ_m) :

The magnetic flux crossing any surface is found by multiplying magnetic flux density with area of the surface, provided that $\vec{B}$ is normal to the surface. If $\vec{B}$ makes some angle with surface not equal to 90° , then

$$\psi_m = \iint_S \vec{B} \cdot d\vec{s} \quad \dots(7.4.2)$$

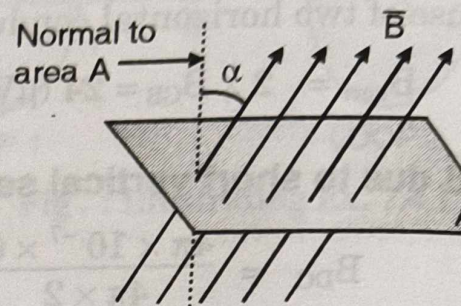


Fig. 7.4.1 : Flux density $\vec{B}$ normal to the area

Where, ψ_m = magnetic flux (Wb)

$\vec{B}$ = Magnetic flux density (Wb/m² or T)

7.4.4 Solved Examples on Magnetic Flux :

Important Formula

$$\psi_m = \iint_S \vec{B} \cdot d\vec{s}$$

Example 7.4.4 : A radial field $\vec{H} = \frac{2.39 \times 10^6}{r} \cos \phi \vec{a}_r$ (A/m)

exists in free space. Find the magnetic flux crossing the surface defined by $0 \leq \phi \leq \pi/4$ and $0 \leq z \leq 1$ m.

Solution : Magnetic flux density is obtained from magnetic field intensity using the relation,

$$\vec{B} = \mu_0 \vec{H} = 4\pi \times 10^{-7} \times \frac{2.39 \times 10^6}{r} \cos \phi \vec{a}_r$$

$$= \frac{3.0}{r} \cos \phi \bar{a}_r \text{ (T)}$$

Magnetic flux crossing the given surface is obtained by $\phi = \iint \bar{B} \cdot d\bar{s}$

For the given surface ϕ and z are varying therefore the third co-ordinate is constant i.e. $r = \text{constant}$. Then,

$$d\bar{s} = r d\phi dz \bar{a}_r$$

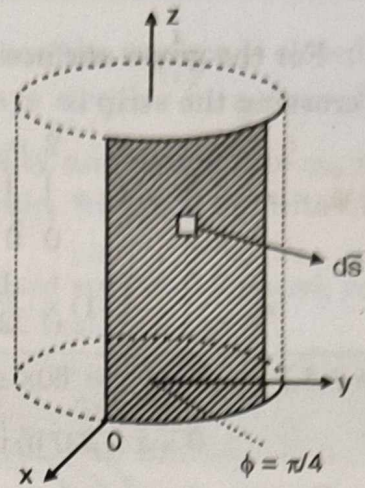


Fig. : Illustrating Example 7.4.4

Thus,

$$\begin{aligned} \psi_m &= \int_0^1 \int_0^{\pi/4} \left(\frac{3.0}{r} \cos \phi \bar{a}_r \right) \cdot (r d\phi dz \bar{a}_r) = 3.0 \times 1 \times \int_0^{\pi/4} \cos \phi d\phi \\ &= 3.0 [\sin \phi]_0^{\pi/4} = 2.12 \text{ (Wb)} \end{aligned}$$

...Ans.

Example 7.4.5 : A radial field $\bar{H} = \frac{2.39 \times 10^6}{r} \cos \phi \bar{a}_r$ (A/m) exits in free space. Find the magnetic flux ϕ crossing the surface defined by $-\pi/4 \leq \phi \leq \pi/4$, $0 \leq z \leq 1$ m.

Tutorial

Solution : Magnetic flux density is obtained from magnetic field intensity using the relation

$$\bar{B} = \mu_0 \bar{H} = 4\pi \times 10^{-7} \times \frac{2.39 \times 10^6}{r} \cos \phi \bar{a}_r = \frac{3.00}{r} \cos \phi \bar{a}_r \text{ (T)}$$

Magnetic flux crossing the given surface is

$$\psi_m = \iint \bar{B} \cdot d\bar{s}$$

For the given surface ϕ and z are varying, therefore the third co-ordinate is constant i.e. $r = \text{constant}$. This means that the differential area is $d\bar{s} = r d\phi dz \bar{a}_r$

$$\begin{aligned} \therefore \psi_m &= \int_0^1 \int_{-\pi/4}^{\pi/4} \left(\frac{3.0}{r} \cos \phi \bar{a}_r \right) \cdot (r d\phi dz \bar{a}_r) = 3.0 \times 1 \times \int_{-\pi/4}^{\pi/4} \cos \phi d\phi \\ &= 3.0 \times [\sin \phi]_{-\pi/4}^{\pi/4} = 4.24 \text{ (Wb)}. \end{aligned}$$

...Ans.

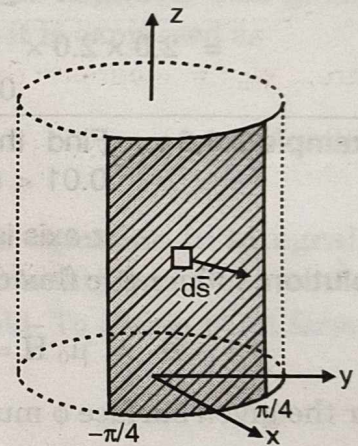


Fig. : Illustrating Example 7.4.5

Example 7.4.6 : Given that $\bar{B} = 2.5 \left(\sin \frac{\pi x}{2} \right) e^{-2y} \bar{a}_z$ Telsa, find the total magnetic flux crossing the strip $z = 0$, $y \geq 0$, $0 \leq x \leq 2$ m.

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Solution : For the given surface $z = \text{constant} = 0$

$$\therefore d\vec{s} = dx dy \vec{a}_z$$

The flux crossing the strip is

$$\begin{aligned} \psi_m &= \int_s \vec{B} \cdot d\vec{s} = \int_0^2 \int_0^\infty 2.5 \left(\sin \frac{\pi x}{2} \right) e^{-2y} dx dy = 2.5 \times \frac{[-\cos(\pi x/2)]_0^2}{\pi/2} \times \left[\frac{e^{-2y}}{-2} \right]_0^\infty \\ &= \frac{2.5}{\pi/2} (-1-1) \times -\frac{1}{2} = 1.59 \text{ Wb} \end{aligned}$$

Example 7.4.7 : Find the flux crossing the plane surface defined by $0.5 \leq r \leq 2.5 \text{ m}$ and $0 \leq z \leq 2.0 \text{ m}$ if $\vec{B} = (20/r) \vec{a}_\phi$ (T).

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Solution : For the given surface ϕ must be constant.

$$\therefore d\vec{s} = dr dz \vec{a}_\phi$$

$$\text{Now, } \psi_m = \int_0^2 \int_{0.5}^{2.5} \vec{B} \cdot d\vec{s} = \int_0^2 \int_{0.5}^{2.5} \left(\frac{2.0}{r} \vec{a}_\phi \right) \cdot (dr dz \vec{a}_\phi)$$

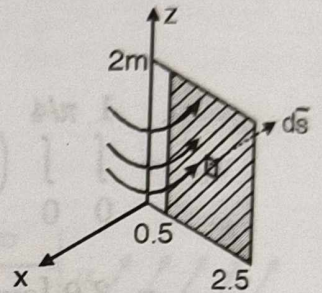


Fig. : Illustrating Example 7.4.

$$= 2.0 \times 2.0 \times \int_{0.5}^{2.5} \frac{dr}{r} = 4 \times [\ln r]_{0.5}^{2.5} = 6.44 \text{ Wb.}$$

...Ans

Example 7.4.8 : Find the flux crossing the portion of the plane $\phi = \pi/4$ defined by $0.01 < r < 0.05 \text{ m}$ and $0 < z < 2 \text{ m}$. A current filament of 2.50 A along the z -axis is in the $\vec{a}_z$ direction.

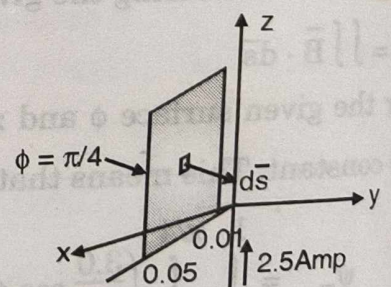
Solution. : Magnetic flux density is obtained by using the relation

$$\vec{B} = \mu_0 \vec{H} = \frac{\mu_0 I}{2\pi r} \vec{a}_\phi$$

For the given surface ϕ must be constant,

$$\therefore d\vec{s} = dr dz \vec{a}_\phi$$

$$\begin{aligned} \text{and } \psi_m &= \int_s \vec{B} \cdot d\vec{s} = \int_0^2 \int_{0.01}^{0.05} \left(\frac{\mu_0 I}{2\pi r} \vec{a}_\phi \right) \cdot (dr dz \vec{a}_\phi) \\ &= \frac{2 \times 4\pi \times 10^{-7} \times 2.5}{2\pi} [\ln r]_{0.01}^{0.05} \\ &= 1.61 \times 10^{-6} \text{ Wb} = 1.61 \mu\text{Wb.} \end{aligned}$$



...Ans. Fig. : Illustrating Example 7.4.8

7.4.5 Gauss's Law for Magnetic Field (Maxwell's Equation) :

The magnetic flux is

$$\psi_m = \int_s \vec{B} \cdot d\vec{s}$$

7.7.1 Solved Examples on Ampere's Circuital Law :

Important Formulae

$$\nabla \times \vec{H} = \vec{J}$$

Expressions for curl in three coordinate systems is given below.

Cartesian system :

$$\nabla \times \vec{H} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix}$$

Cylindrical system :

$$\nabla \times \vec{H} = \begin{vmatrix} \frac{1}{r}\vec{a}_r & \vec{a}_\phi & \frac{1}{r}\vec{a}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ H_r & rH_\phi & H_z \end{vmatrix}$$

Spherical system :

$$\nabla \times \vec{H} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \vec{a}_r & r\vec{a}_\theta & r \sin \theta \vec{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ H_r & rH_\theta & r \sin \theta H_\phi \end{vmatrix}$$

If you find these three equations difficult to remember, you can use method explained in Chapter 1.

Example 7.7.1 : An $\vec{H}$ due to a current source is given by $\vec{H} = y \cos(2x) \vec{a}_x + (y + e^x) \vec{a}_z$. Describe the current density at origin. MU - May 2005

Solution : We have $\vec{J} = \nabla \times \vec{H}$

In cartesian co-ordinate system

$$\begin{aligned} \vec{J} &= \nabla \times \vec{H} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y \cos 2x & 0 & y + e^x \end{vmatrix} \\ &= \vec{a}_x (1) - \vec{a}_y (e^x) + \vec{a}_z (-\cos 2x) = \vec{a}_x - e^x \vec{a}_y - \cos 2x \vec{a}_z \end{aligned}$$

At origin (0,0,0)

$$\vec{J} = \vec{a}_x - e^0 \vec{a}_y - \cos(0) \vec{a}_z = \vec{a}_x - \vec{a}_y - \vec{a}_z \text{ (A/m}^2\text{)}$$

...Ans.

Example 7.7.2 : In a certain conducting region
 $\vec{H} = yz(x^2 + y^2) \vec{a}_x - y^2 xz \vec{a}_y + 4x^2 y^2 \vec{a}_z$ (A/m)

i) Determine $\vec{J}$ at (5, 2, -3)

ii) Show that $\nabla \cdot \vec{B} = 0$

Solution :

i) To find $\bar{J}$:

$$\begin{aligned}\bar{J} &= \nabla \times \bar{H} = \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz(x^2+y^2) & -y^2xz & 4x^2y^2 \end{vmatrix} \\ &= \bar{a}_x \left[\frac{\partial}{\partial y} (4x^2y^2) - \frac{\partial}{\partial z} (-y^2xz) \right] - \bar{a}_y \left[\frac{\partial}{\partial x} (4x^2y^2) - \frac{\partial}{\partial z} yz(x^2+y^2) \right] \\ &\quad + \bar{a}_z \left[\frac{\partial}{\partial x} (-y^2xz) - \frac{\partial}{\partial y} yz(x^2+y^2) \right] \\ &= [8x^2y + xy^2] \bar{a}_x - [y(x^2+y^2) - 8xy^2] \bar{a}_y + [y^2z + z(x^2+3y^2)] \bar{a}_z\end{aligned}$$

$$\begin{aligned}\bar{J}|_{(5,2,-3)} &= [8(5^2)(2) + 5(2)^2] \bar{a}_x - [2(5^2 + 2^2) - 8(5)(2^2)] \bar{a}_y - [(2^2)(-3) + (-3)(5^2 + 3(2^2))] \bar{a}_z \\ &= 420 \bar{a}_x - 102 \bar{a}_y + 123 \bar{a}_z\end{aligned}$$

...Ans.

ii) To show that $\nabla \cdot \bar{B} = 0$

$$\begin{aligned}\nabla \cdot \bar{B} &= \mu_0 (\nabla \cdot \bar{H}) = \mu_0 \left[\frac{\partial}{\partial x} yz(x^2+y^2) + \frac{\partial}{\partial y} (-y^2xz) + \frac{\partial}{\partial z} (4x^2y^2) \right] \\ &= \mu_0 [2xyz - 2yxz + 0] = 0\end{aligned}$$

...Ans.

Hence proved.

Example 7.7.3 : Find a current distribution that produces a magnetic field of the form $\bar{H} = k \sin x \bar{a}_y$, where k is a constant.

Solution : The current density that produces the magnetic field of the given form can be obtained by using.

$$\begin{aligned}\bar{J} &= \nabla \times \bar{H} = \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ H_x & H_y & H_z \end{vmatrix} = \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 0 & k \sin x & 0 \end{vmatrix} \\ &= \bar{a}_x (0) - \bar{a}_y (0) + \bar{a}_z \frac{\partial}{\partial x} (k \sin x) = k \cos x \bar{a}_z \text{ (A/m}^2\text{)}\end{aligned}$$

...Ans.

Example 7.7.4 : If magnetic field intensity in a region is

$$\bar{H} = x^2 \bar{a}_x + 2yz \bar{a}_y + (-x^2) \bar{a}_z$$

Find the current density at point (i) (2, 3, 4) (ii) (1, 1, -1) and (0, 0, 0).

Solution : We know that, point form of Ampere's law is

$$\bar{J} = \nabla \times \bar{H} = \begin{vmatrix} \bar{a}_x & \bar{a}_y & \bar{a}_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 & 2yz & -x^2 \end{vmatrix}$$

So, $\nabla \times \vec{H} = (0 - 2y) \vec{a}_x - (-2x - 0) \vec{a}_y + (0 - 0) \vec{a}_z$

$\therefore \vec{J} = -2y \vec{a}_x + 2x \vec{a}_y \text{ (A/m}^2\text{)}$

Current density $\vec{J}$ at :

(i) $(2,3,4) \vec{J} = -2 \times (3) \vec{a}_x + 2(2) \vec{a}_y$ i.e. $\vec{J} = -6 \vec{a}_x + 4 \vec{a}_y \text{ (A/m}^2\text{)}$

...Ans.

(ii) $(1,1,-1) = -2 \times (1) \vec{a}_x + 2(1) \vec{a}_y = -2 \vec{a}_x + 2 \vec{a}_y \text{ (A/m}^2\text{)}$

...Ans.

(iii) $(0,0,0) = -2 \times (0) \vec{a}_x + 2(0) \vec{a}_y = 0$

...Ans.

Example 7.7.5 : Find the current I through a square area 2m on a side with edges coinciding with $+x$ and $+y$ axes and one corner at the origin if

(i) $\vec{H} = 2y^2 \vec{a}_x \text{ (A/m)}$

(ii) $\vec{H} = 3x^2y \vec{a}_z \text{ (A/m)}$

Solution : Using point form of Ampere's law

(i) $\vec{J} = \nabla \times \vec{H} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2y^2 & 0 & 0 \end{vmatrix} = -4y \vec{a}_z$

Now, $I = \int_S \vec{J} \cdot d\vec{s}$

For a given area $d\vec{s} = dx dy \vec{a}_z$

$I = \int_0^2 \int_0^2 (-4y \vec{a}_z) \cdot (dx dy \vec{a}_z) = \int_0^2 \int_0^2 -4y dx dy = -16 \text{ Amp.}$

...Ans.

(ii) $\vec{J} = \nabla \times \vec{H} = \begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 0 & 0 & 3x^2y \end{vmatrix} = \vec{a}_x (3x^2) - \vec{a}_y (6xy)$

$\therefore I = \int_S \vec{J} \cdot d\vec{s}$

$= \int_0^2 \int_0^2 (3x^2 \vec{a}_x - 6xy \vec{a}_y) \cdot dx dy \vec{a}_z = 0 \text{ Amp} \dots (\because \vec{a}_y \cdot \vec{a}_z = 0)$

...Ans.